

Report: Order Delivery Time Prediction

Objectives

The objective of this assignment is to build a regression model that predicts the delivery time for orders placed through Porter. The model will use various features such as the items ordered, the restaurant location, the order protocol, and the availability of delivery partners.

The key goals are:

- Predict the delivery time for an order based on multiple input features
- Improve delivery time predictions to optimise operational efficiency
- Understand the key factors influencing delivery time to enhance the model's accuracy

Below are the Steps taken:

1. Data Preparation:

- Fixing the Datatypes : Converted the date and time fields to appropriate datatype, which is date time using pandas library.
- Also converted Categorical fields to type Category.

2. Feature Engineering:

- Calculated the time taken using the features `actual_delivery_time` and `created_at`, which is calculated by `(actual_delivery_time - created_at)`, which is also the predictor variable.
- Extracted the hour of the day from the `created_at` timestamp to capture time-of-day patterns in ordering behaviour.
- Removed irrelevant columns like order IDs and original timestamps to avoid data leakage and redundancy.

3. Creating Training and Validation sets:

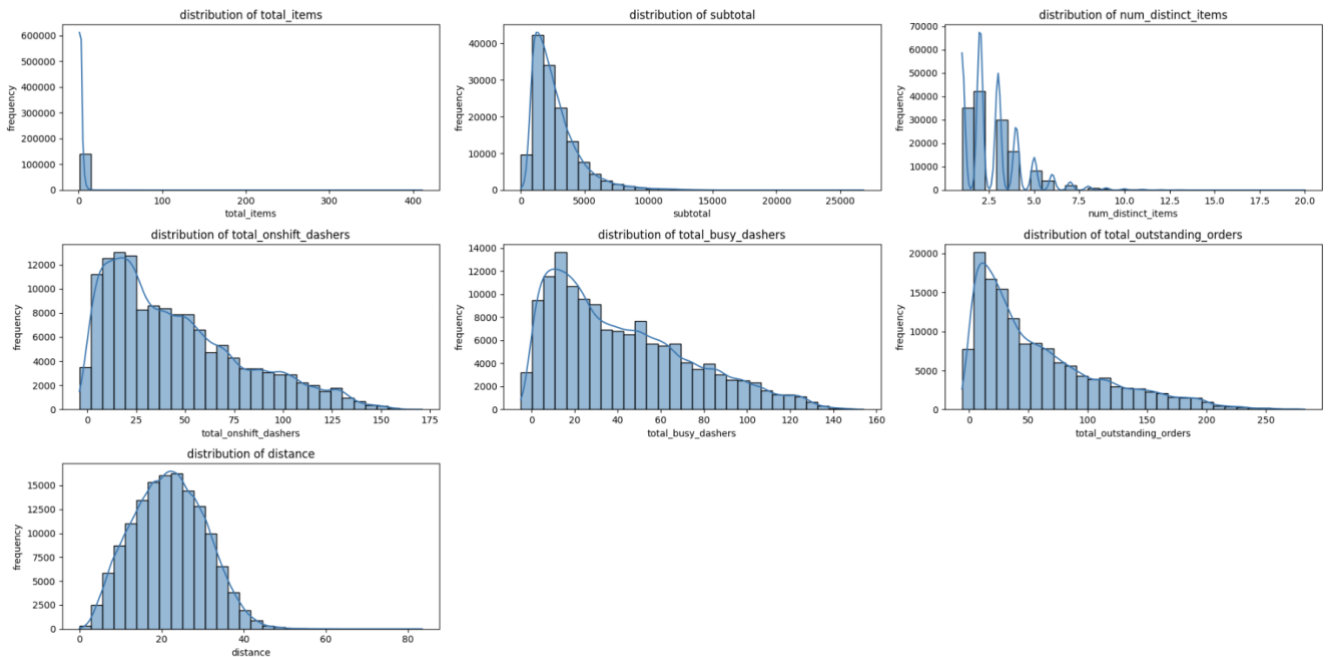
- The dataset was split into training and testing subsets using an 80-20 ratio. Further split into `X_train`, `y_train`, `X_test`, `y_test`. This step ensured that model performance could be evaluated on unseen data.

4. Exploratory Data Analysis on Training Data:

- Firstly, I defined the numerical and categorical features in the dataset. This was essential for organizing the data and making it easier to apply the appropriate techniques for analysis.

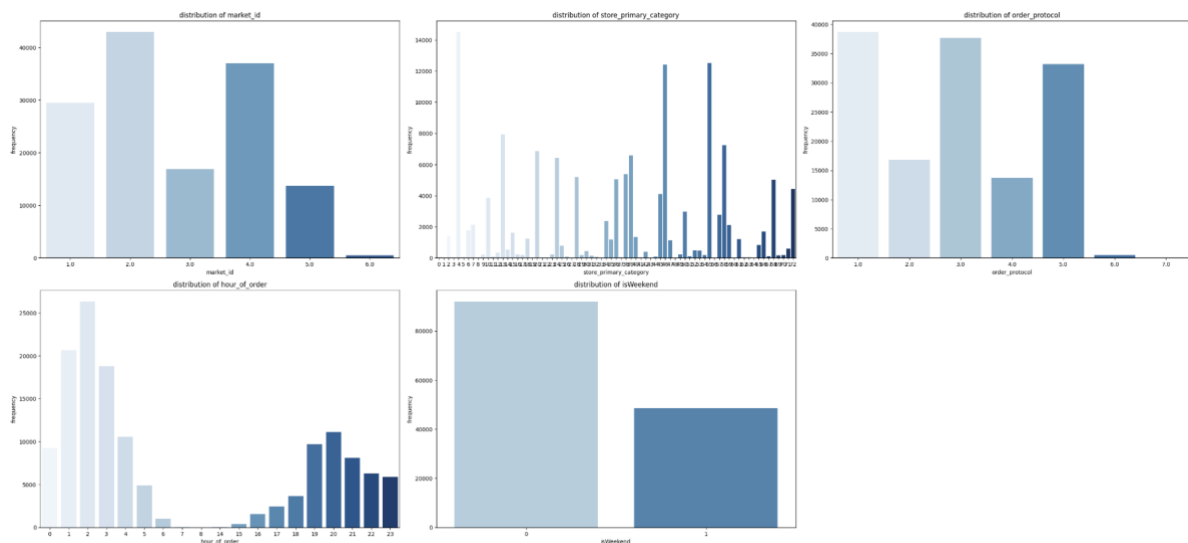
1. Visualizing the Distribution of Numerical Features:

- I created histograms with KDE to understand the distribution of the numerical features. These visualizations helped in identifying the central tendency and spread of each feature.



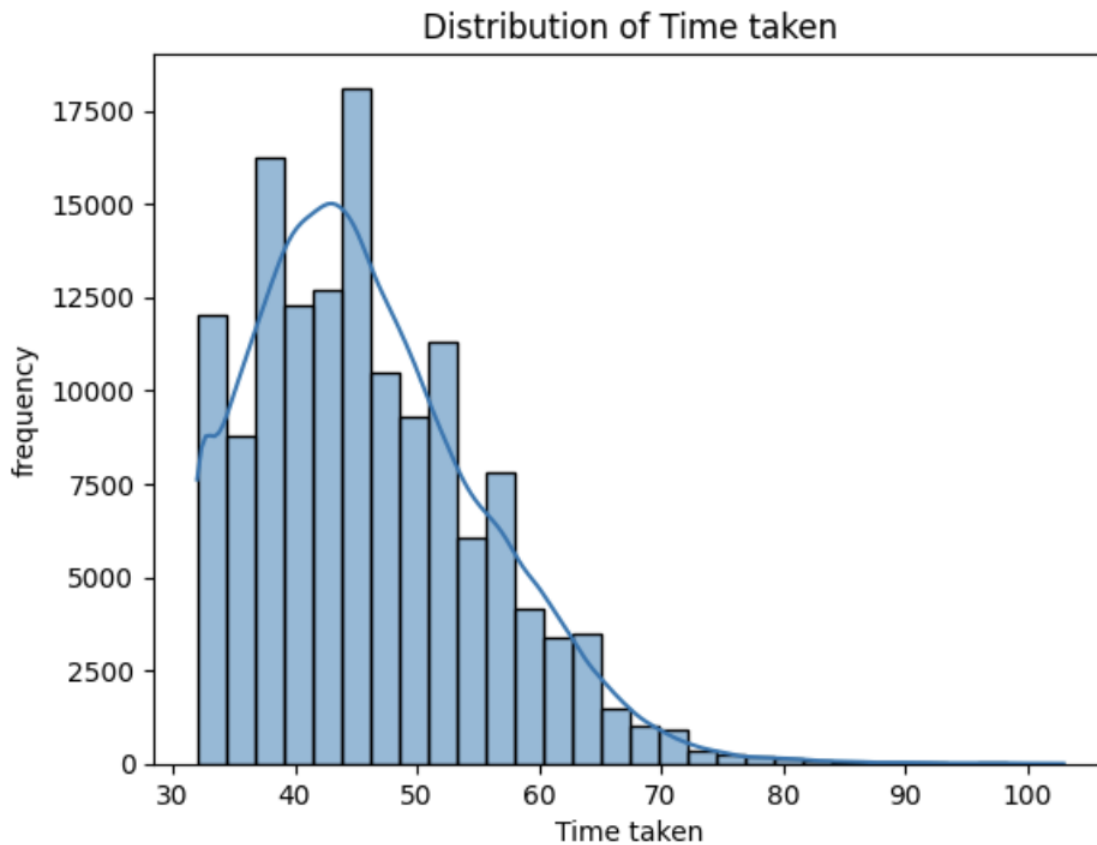
2. Visualizing the Distribution of Categorical Features:

- I used count plots to check the distribution of the categorical variables. This helped me understand how often each category appeared in the dataset and whether there were any imbalances.

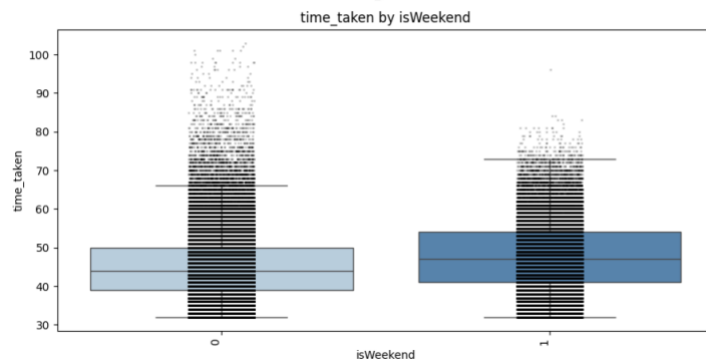
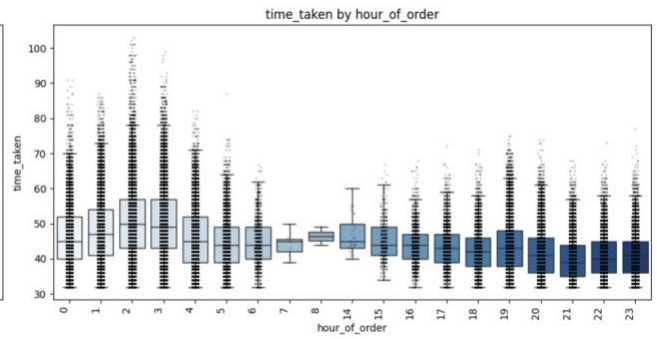
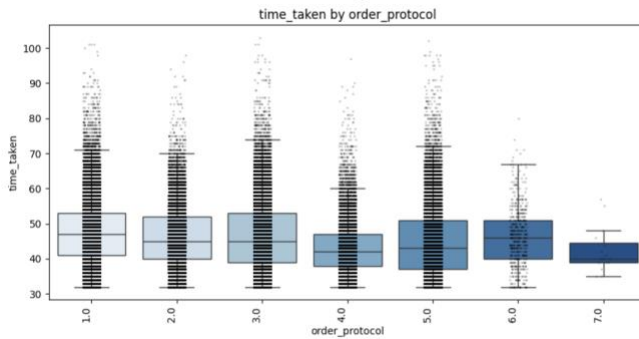
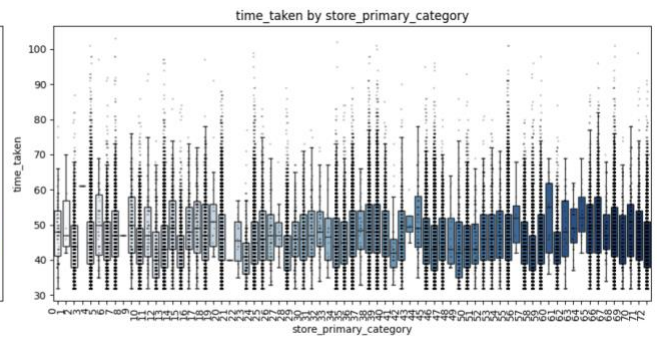
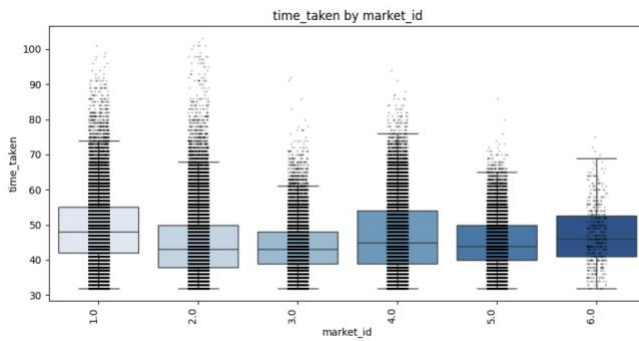
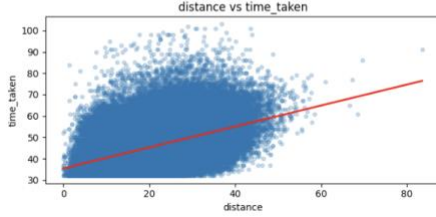
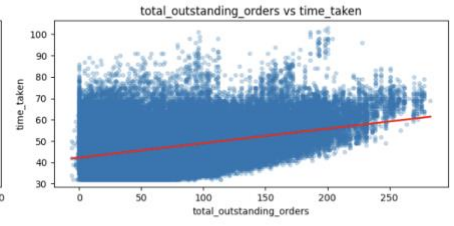
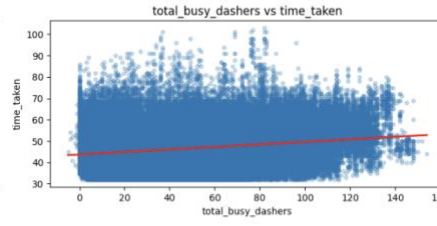
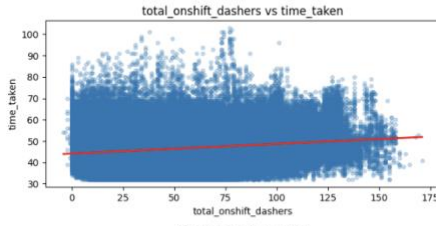
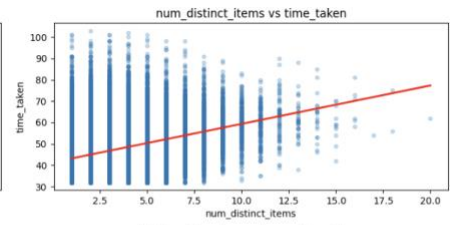
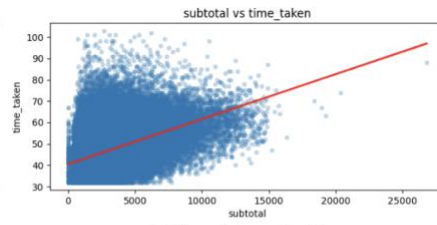
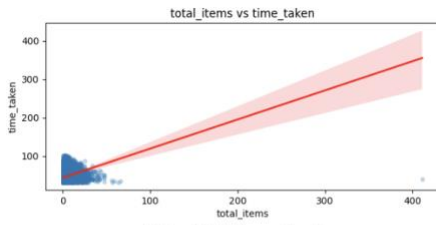


3. Distribution of Time Taken (Target Variable):

- Next, I visualized the distribution of the target variable, time_taken, to understand how delivery times were spread out. The histogram and KDE plot revealed that time_taken was slightly skewed.

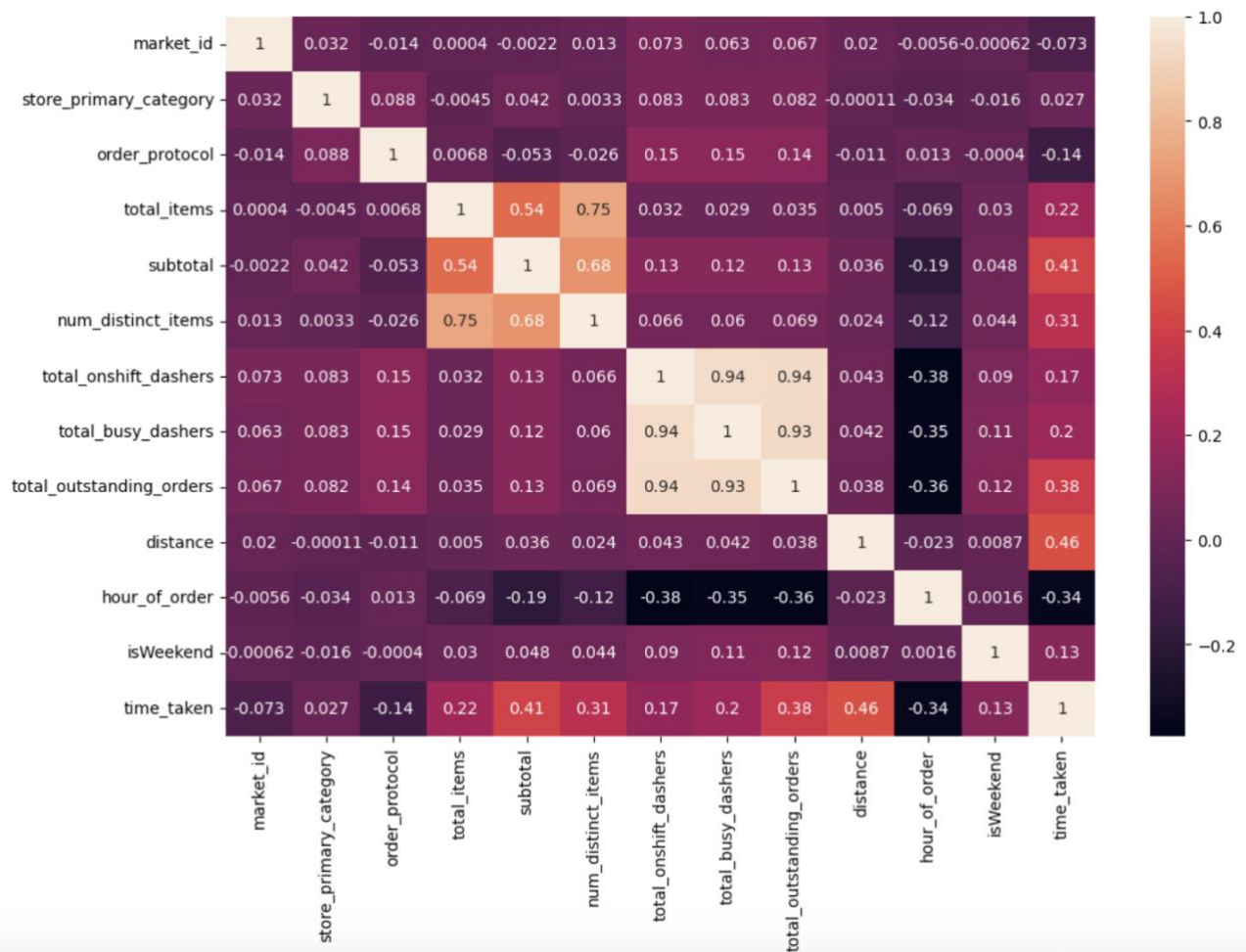


- Next I explored how each feature relates to time_taken by plotting both regression plots for numerical features and box plots for categorical features. This helped me visually assess how each predictor impacted delivery time.



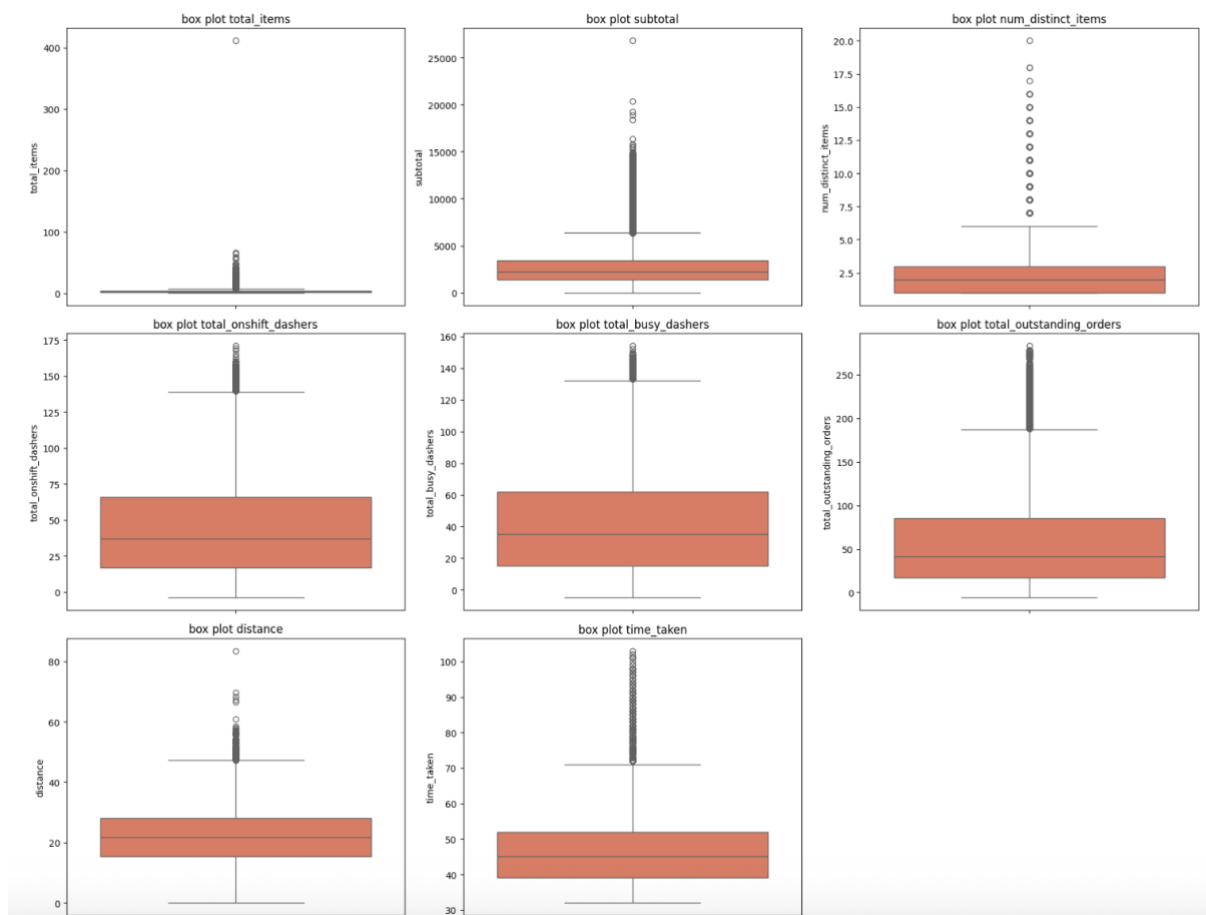
5. Correlation Analysis:

- To further explore relationships between the features, I created a correlation heatmap, to identify which features have strong correlations with time_taken.
- Feature having the strongest correlation is distance(0.46), and weak features are market_id, store_primary_category etc, hence dropped these features.



6. Handling Outliers:

- Outliers were detected by using box plots. Outliers can distort model performance, so I decided to cap the values at the 99th percentile.



7. EDA was also performed on the test dataset to verify if it behaves similarly to the training data. The same columns were dropped from the test dataset, and outliers were capped using the 99th percentile values derived from the training data to prevent data leakage.

8. Model Building:

1. Feature Scaling:

- 1.1.** MinMaxScaler is used to scale the numerical features. The MinMaxScaler transforms each feature so that it falls within a specified range (typically between 0 and 1). This ensures that all features contribute equally to the model, avoiding the situation where features with larger values dominate the learning process.
- 1.2.** angle_train is created by converting the hour_of_order (which is cyclic) into a sine and cosine transformation to retain the cyclical nature of time. This transformation helps the model understand time-related patterns (e.g., 11 AM and 1 PM being closer to each other).
- 1.3.** After creating hour_sin and hour_cos, the original hour_of_order feature is dropped to prevent redundancy and maintain a leaner model.
- 1.4.** Scaling is applied to numerical variables using the MinMaxScaler. This process scales the values of features like total_items, subtotal, etc., into a range from 0 to 1, so they are treated equally by the model, especially since the scale of some features can be quite different.

2. Build a Linear Regression Model:

- 2.1. Preparing the Test Dataset: Before building the model, I applied the same transformations to the test data as I did for the training data. This ensures that the features are consistent across both datasets.
- 2.2. I created a Linear Regression model using the training data. The model was trained on the scaled and transformed features to predict delivery time.
- 2.3. Once the model was trained, I used it to predict delivery times for both the training and test datasets (y_{train_pred} and y_{test_pred}).

2.4. Evaluation Metrics:

- 2.4.1. MAE (Mean Absolute Error): Average magnitude of errors in the predictions, irrespective of their direction. Training and Test MAE are 2.7 and 2.68 minutes respectively.
- 2.4.2. RMSE (Root Mean Squared Error): This gives a measure of the error based on the square of the differences. Training RMSE is 3.66 minutes and Test RMSE is 3.64 minutes.
- 2.4.3. R-Squared: explains the proportion of the variance in the target variable that can be explained by the model, which ranges between 0 to 1, where 1 being perfect prediction. So training R-Squared is .845 and test R-Squared is .848.
- 2.4.4. Adjusted R-Squared adjusts the R-Squared value by taking the number of features into account, which is useful when comparing models with different numbers of predictors, values for training and test are .846 and .848 respectively.

3. All coefficients have p-values close to 0, meaning they are statistically significant. It indicates that each of these features plays a significant role in predicting the delivery time.

- 3.1. Features like subtotal, distance, and total_outstanding_orders have a strong positive impact on delivery time, which is consistent with expectations in a delivery scenario.
- 3.2. The negative coefficient for total_items could be surprising because more items typically mean more time needed for delivery. This could be a result of correlations with other variables or a unique dataset trend.

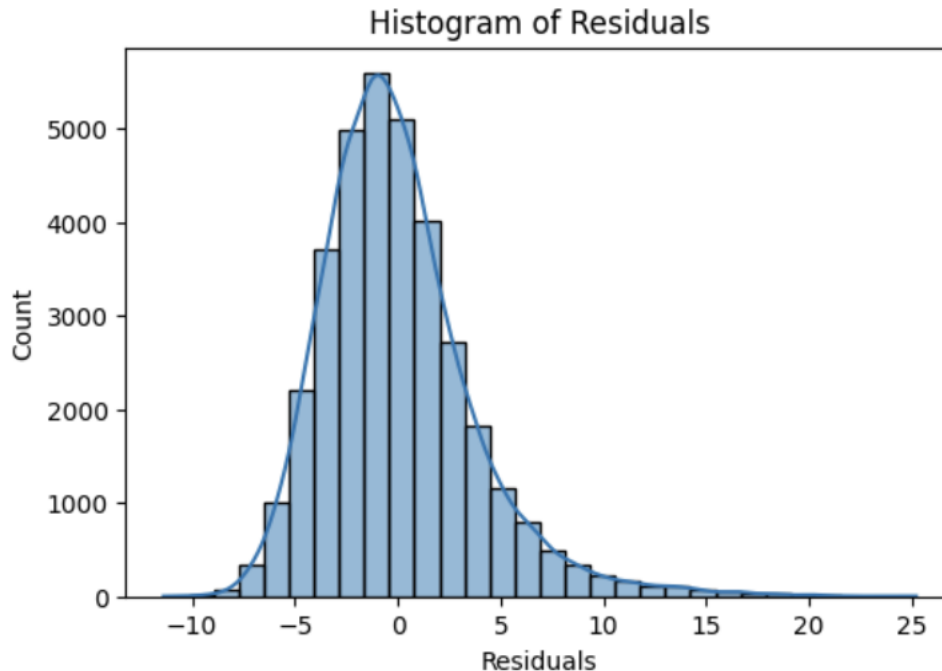
9. Recursive Feature Elimination (RFE):

The idea behind RFE is to recursively remove features, one by one, and evaluate the model performance at each iteration, keeping the features that provide the best performance.

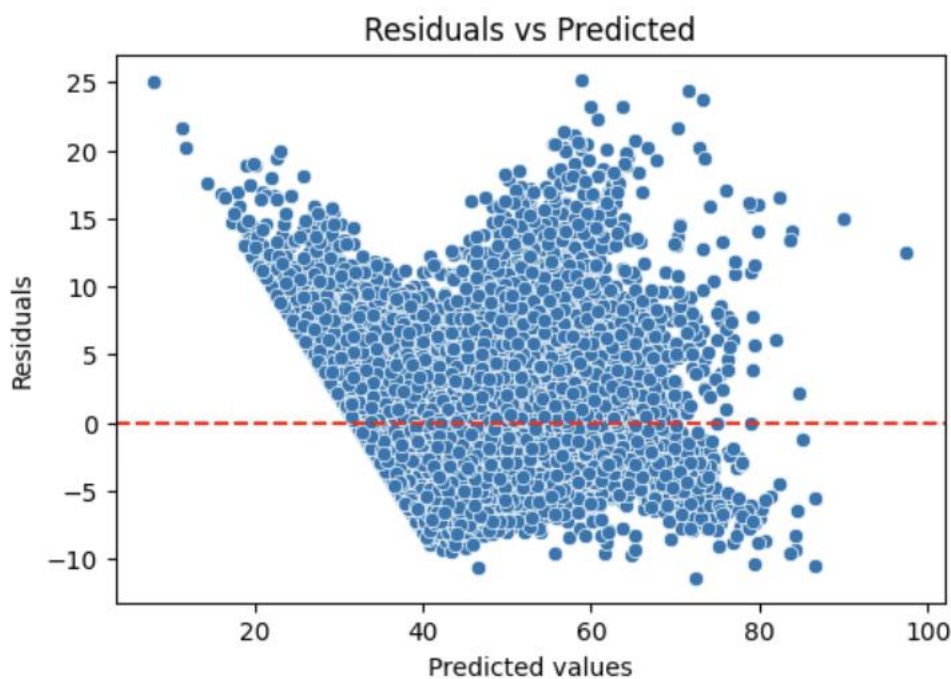
- 9.1. I started with all features and used Linear Regression as the estimator in RFE. The number of features to be selected started from the total available features, and gradually reduced from 10 down to 7.
- 9.2. After analysing the performance metrics (R-Squared and RMSE), I selected the top 7 features based on a balance of performance and feature count.
- 9.3. The model was trained and evaluated on both the training and test datasets using selected 7 features.

10. Residual Analysis: Residual plots help us check if the assumptions of linear regression hold for our model.

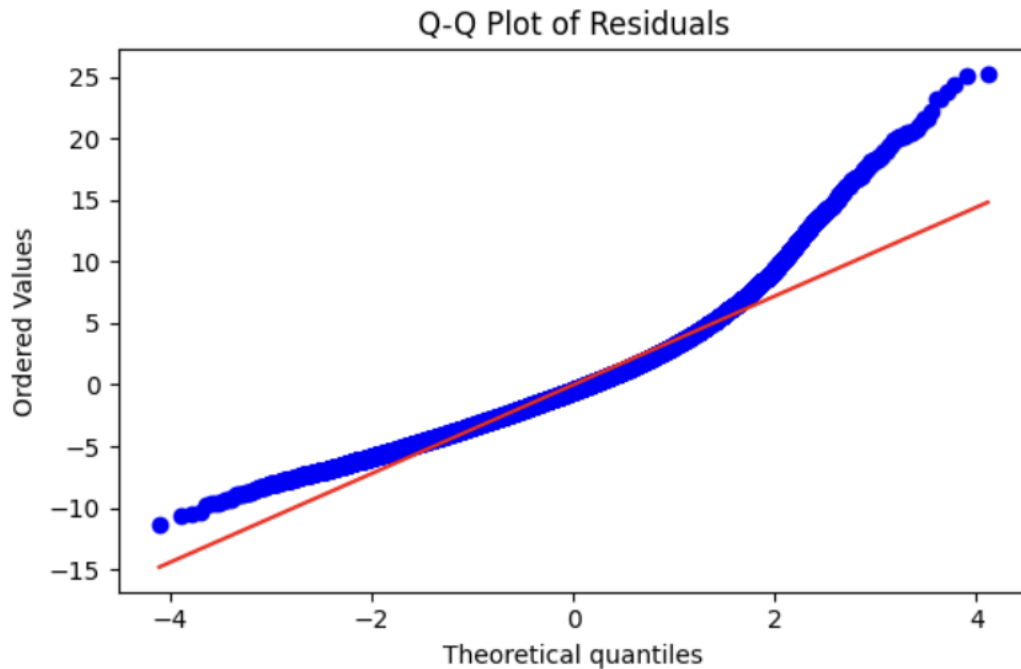
10.1. Histogram of Residuals: This shows the distribution of errors between predicted and actual values. The residuals are mostly cantered around zero but show a slight right skew, indicating some errors tend to be positive and larger. This suggests residuals are not perfectly normal but close.



10.2. Residuals vs Predicted Values: This plot checks if residuals have constant variance (homoscedasticity). Ideally, residuals scatter randomly around zero. Here, a funnel shape appears, indicating heteroscedasticity - variance increases as predictions grow, which means model errors grow with larger delivery times.



- 10.3. Q-Q Plot of Residuals:** This compares residual quantiles to a normal distribution. Points follow a roughly straight line but diverge in tails, confirming the residuals are not perfectly normal, supporting the slight skew seen in the histogram.



- 11.** The final linear regression model, built using 7 key features, is able to explain about 84% of the variation in delivery times and keeps the MAE around 2.8 min. The error is also stable across both training and test data, which means the model is reliable and not overfitting.

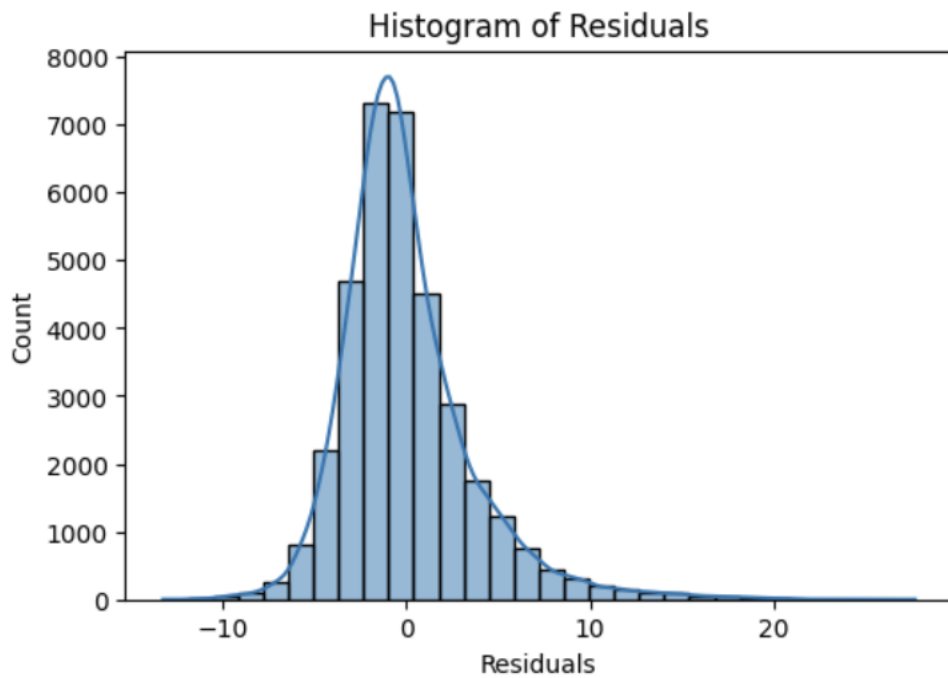
While reviewing the model, I found a high VIF between some dasher-related features, like total dashers on shift and those already busy. These features are highly multicollinear, which doesn't affect the overall prediction accuracy but makes it harder to interpret the individual importance of each feature.

Overall, the model is well-suited for predicting delivery times in real-world operations and gives dependable results.

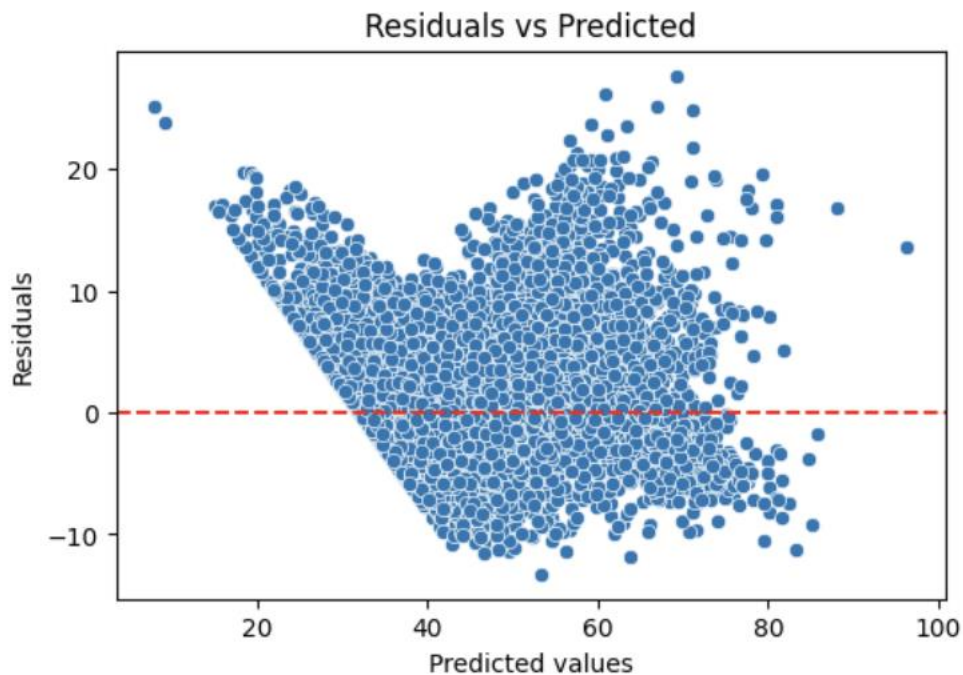
12. Multicollinearity Handling and Model Refinement:

- 12.1.** Started with all selected features, the VIF values revealed high multicollinearity among total_busy_dashers, total_onshift_dashers, and total_outstanding_orders - all with VIF > 20, which indicates strong correlation among these predictors.
- 12.2.** To reduce multicollinearity, I sequentially dropped: total_busy_dashers, after removal, model R-Squared dropped slightly but remained strong. num_distinct_items removal led to a slight decrease in R-Squared 0.849, still acceptable.
- 12.3. Trade-Off:** Removing highly collinear features lowered VIF values significantly but attempting to drop any further important features (especially total_onshift_dashers or total_outstanding_orders) caused dramatic drop in R-Squared around .50, indicating loss of predictive power.
- 12.4.** Based on this, I retained the features with high VIF but strong predictive importance (total_onshift_dashers, total_outstanding_orders) and removed only total_busy_dashers and num_distinct_items, which is a practical compromise balancing model interpretability and accuracy.

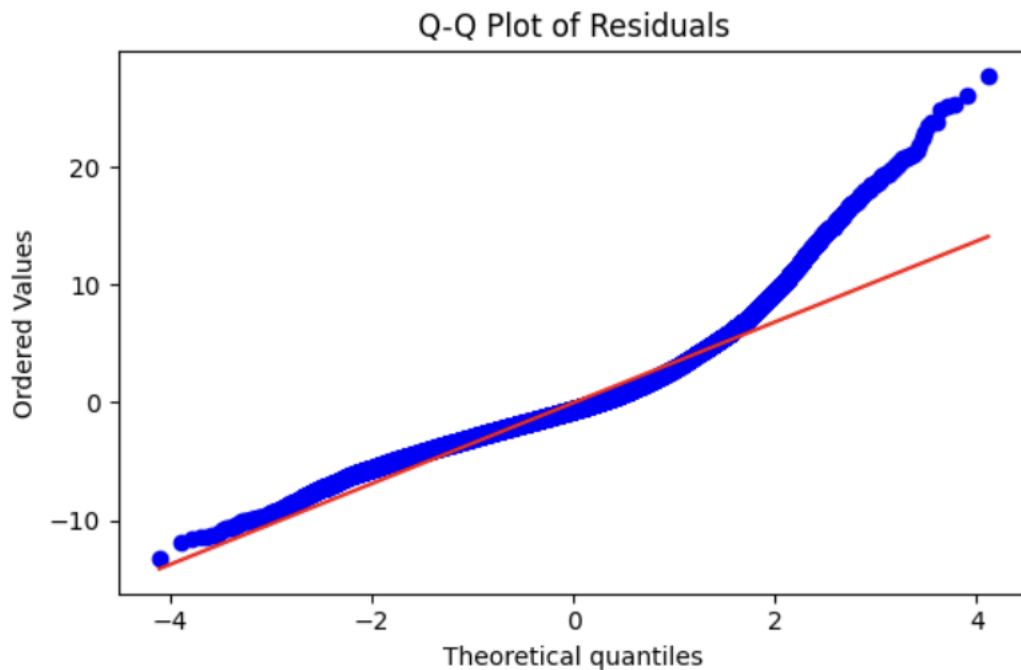
- 13.** The overall shape of the residual distribution remains similar to the previous analysis. However, after feature adjustments, the residuals show a bit tighter spread around zero, suggesting improved model stability.



- 13.1.** The residuals vs. predicted plot still shows increasing variance with higher predicted values, which shows heteroscedasticity remains in the improved Model.



- 13.2.** The Q-Q plot continues to demonstrate approximate normality of residuals, with deviations mostly at the extremes, consistent with earlier findings.



14. Coefficients Analysis: The model coefficients indicate how each feature influences the delivery time. Since most features were scaled during preprocessing, direct interpretation of these scaled coefficients can be misleading. To better understand the real-world impact, I calculated the unit change impact, which shows how much the delivery time changes for a one-unit increase in each unscaled feature.

14.1. Total Items: Each additional item in an order increases the delivery time by approximately 0.004 minutes (about 0.24 seconds).

14.2. Subtotal: A one-unit increase in subtotal (meaning 1 rupee) has a very small effect (~0.0006 minutes), which accumulates with larger changes.

14.3. Total OnShift Dashers: Increasing dashers on shift by one reduces delivery time by roughly 0.003 minutes (about 0.18 seconds), reflecting improved delivery capacity.

14.4. Total Outstanding Orders: Each additional pending order increases delivery time by about 0.24 minutes, indicating workload impact.

14.5. Distance: Every additional km increases delivery time by approximately 0.24 minutes, consistent with expectations.

	Feature	Scaled Coef	Range	Unit_change_impact
1	total_items	1.665333	410.00	0.0041
2	subtotal	15.198113	26800.00	0.0006
3	total_onshift_dashers	-0.453631	175.00	-0.0026
4	total_outstanding_orders	71.091530	291.00	0.2443
5	distance	19.989915	83.52	0.2393